

A Literature-Implied Sharpening of the Complex Median Lemma in arXiv:2407.14988

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Status. `literature_implied_answer (complex median lemma sharpening; not main open problem)`. This note records a scoped identification: the complex median lemma in Feng–Wang–Zhou [1] is a pushforward-measure instance of the planar orthogonal equipartition or second pancake theorem. The identification improves the lemma’s constant from $1/16$ to the optimal $1/4$. It does not solve the paper’s open noncommutative paraproduct boundedness problem, nor the low- p endpoint range left open for Theorem 1.6 when $n = 1$.

Source question/passage. Theorem 1.9 of [1] (TeX label `divideS`, proved in Section 8) states that if $(\Omega, \mathcal{F}, \mu)$ is a measure space, $I \in \mathcal{F}$ has finite measure, and $b : I \rightarrow \mathbb{C}$ is measurable, then there are two orthogonal lines L_1, L_2 in $\mathbb{C}$ whose four closed quadrants $T_1, \dots, T_4$ satisfy

$$\mu(\{x \in I : b(x) \in T_i\}) \geq \frac{1}{16}\mu(I), \quad i = 1, \dots, 4.$$

After the proof, the source remarks that proving the theorem first for simple functions and then passing to general functions by a limit argument is not clear.

Supporting theorem. Fakhrutdinov–Musin [2] recall the planar second pancake theorem: a finite-area planar mass admits two perpendicular straight lines cutting the mass into four equal pieces. Their Theorem 1 proves the finite-point version algorithmically. The measure form is the exact geometric statement needed after pushing $\mu|_I$ forward under b .

Theorem 1 (Optimal closed-quadrant complex median). *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, let $I \in \mathcal{F}$ have finite measure, and let $b : I \rightarrow \mathbb{C}$ be measurable. Then there exist two perpendicular lines $L_1, L_2 \subset \mathbb{C}$ such that, if $Q_1, \dots, Q_4$ are the four closed quadrants determined by these lines, then*

$$\mu(\{x \in I : b(x) \in Q_j\}) \geq \frac{1}{4}\mu(I), \quad j = 1, \dots, 4.$$

The constant $1/4$ is the best possible universal constant.

Proof. Let $M = \mu(I)$. If $M = 0$ there is nothing to prove. Put

$$\nu = b_{\#}(\mu|_I),$$

a finite Borel measure on $\mathbb{C} \simeq \mathbb{R}^2$ with total mass M . It is enough to find the perpendicular cuts for ν and then pull them back through b .

For compactly supported absolutely continuous planar measures, the second pancake theorem gives perpendicular lines whose four quadrants have exactly mass $M/4$. For a general finite Borel

measure ν , choose compactly supported absolutely continuous measures ν_m with $\nu_m(\mathbb{C}) = M$ and $\nu_m \Rightarrow \nu$ weakly. Let $L_{1,m}, L_{2,m}$ be perpendicular exact-quarter cuts for ν_m .

The family of cuts has a convergent subsequence. Indeed, by tightness choose a disk B_R with $\nu_m(B_R) > M/2$ for all large m . Each of the two cutting lines is a median line for ν_m , since the two adjacent quadrants on either side have total mass $M/2$. Hence neither line can miss B_R ; otherwise one of its halfplanes would have mass $< M/2$. Thus the offsets of both lines are bounded, and since the two lines are perpendicular their intersection points and directions lie in a compact parameter set. Passing to a subsequence, the two lines converge to perpendicular lines L_1, L_2 .

Let Q_j be a closed quadrant for the limiting pair, and let $Q_{j,m}$ be the corresponding closed quadrant for the approximating pair. For every $\delta > 0$, eventually $Q_{j,m} \subset (Q_j)_\delta$, the closed δ -neighborhood of Q_j . Therefore

$$\frac{M}{4} = \nu_m(Q_{j,m}) \leq \nu_m((Q_j)_\delta).$$

Since $(Q_j)_\delta$ is closed and $\nu_m \Rightarrow \nu$, the Portmanteau theorem gives

$$\frac{M}{4} \leq \nu((Q_j)_\delta).$$

Letting $\delta \downarrow 0$ and using continuity from above for finite measures yields $\nu(Q_j) \geq M/4$. Pulling back by b gives the asserted inequality.

Optimality follows by applying any stronger putative constant to a line-null planar mass, such as normalized area measure on a disk. The four open quadrants are disjoint up to null boundary, so no universal lower bound can exceed $1/4$. $\square$

Scope and novelty. The supporting authors do not present this as an answer to arXiv:2407.14988. The implication is obtained by identifying the source’s complex-valued median construction with the classical orthogonal equipartition theorem for the pushforward measure. The result is useful as a cleaner and sharper replacement for Theorem 1.9, and it gives the source’s desired limit argument in the closed-quadrant formulation. It should not be counted as progress on the main operator-valued open questions in the source paper.

References

- [1] X. Feng, T. Wang, and Y. Zhou, *On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators*, arXiv:2407.14988, 2024. <https://arxiv.org/abs/2407.14988>
- [2] A. Fakhrutdinov and O. R. Musin, *Algorithms for orthogonal partitioning into four parts*, arXiv:2511.20866v2, 2026. <https://arxiv.org/abs/2511.20866>